

Midterm Exam 1 Sample

ECSE 4560/6560: Modern Communication Systems

RPI Fall 2025

Question 1: Wireless Transceiver Chain and Modulation

- (a) Explain why *up-conversion* is necessary in wireless systems. What fundamental limitation would arise if we tried to transmit at baseband frequencies?
- (b) For a system operating at carrier frequency $f_c = 2.4$ GHz and bandwidth $B = 20$ MHz, what is the minimum sampling rate required to directly digitize the passband signal? Compare it with the rate required for complex baseband sampling.
- (c) A transmitter uses 16-QAM with average symbol energy $E_s = 10^{-3}$ J. Sketch the constellation and compute the minimum Euclidean distance between points.
- (d) If the same transmit power is used for BPSK, QPSK, and 16-QAM, explain qualitatively which modulation achieves the lowest BER and why.

Question 2: Estimation and Decision Theory

- (a) Consider a binary detection problem $y = s + n$, where $s \in \{+A, -A\}$ and $n \sim \mathcal{N}(0, \sigma^2)$. Derive the **Maximum Likelihood (ML)** decision rule and the corresponding decision boundary.
- (b) Modify the result for the **MAP detector** if $P(s = +A) = 0.8$ and $P(s = -A) = 0.2$. What happens to the threshold?
- (c) Using the Q -function, derive the expression for bit-error probability

$$P_b = Q\left(\sqrt{\frac{2E_b}{N_0}}\right)$$

for BPSK over AWGN.

- (d) How would the performance change if we use 16-QAM instead of BPSK at the same E_b/N_0 ? Explain using constellation geometry and SNR per bit reasoning.

Question 3: Sampling and Reconstruction

- (a) State the **Nyquist Sampling Theorem**. Explain what happens when the sampling rate is below, equal to, and above $2B$, where B is the signal bandwidth.
- (b) A continuous-time baseband signal $x(t)$ with maximum frequency 3 MHz is sampled at 4 MHz. Sketch its sampled spectrum and explain the aliasing that occurs.
- (c) Describe ideal reconstruction using an LPF. Provide the expression for the reconstructed signal in terms of sinc interpolation.

Question 4: Wireless Channel Effects

- (a) Express the general received signal model $y(t) = h(t) * x(t) + v(t)$. List and describe the four key effects introduced by the channel: noise, delay, attenuation, and phase rotation.
- (b) Using the free-space path loss formula, compute the received power in dBm at 10 m for $f = 5$ GHz and $P_T = 0$ dBm. Assume Omni-directional antennas.
- (c) Explain and illustrate multipath fading with two propagation paths of lengths 1 m and 1.06 m at 5 GHz. Calculate the phase difference and discuss constructive vs. destructive interference.
- (d) Define channel impulse response (CIR) and show how it leads to inter-symbol interference (ISI).
- (e) Describe a simple preamble-based channel estimation and equalization scheme for BPSK and explain how it restores coherence at the receiver.

Question 5: Impact of Multipath on OFDM Orthogonality

An OFDM system uses $N = 64$ subcarriers and a sampling rate of $f_s = 20$ MHz. The cyclic prefix (CP) length is $N_{CP} = 16$ samples. The channel has two multipath components given by:

$$h(t) = \delta(t) + 0.8 e^{-j\pi/3} \delta(t - \tau_d),$$

where the delay spread is $\tau_d = 0.9 \mu\text{s}$.

- (a) Compute the OFDM symbol duration *with* and *without* CP in μs , and determine the corresponding subcarrier spacing in kHz. Briefly explain how delay spread relates to frequency selectivity in wireless channels.
- (b) Determine whether the given delay spread τ_d fits within the CP duration. If not, estimate how many time-domain samples of the delayed path extend beyond the CP. Discuss qualitatively how this affects subcarrier orthogonality and inter-symbol interference (ISI).
- (c) Write the frequency-domain response of the two-tap channel for subcarrier k :

$$H(f_k) = 1 + 0.8 e^{-j(2\pi f_k \tau_d + \pi/3)}.$$

Sketch the magnitude of $|H(f_k)|$ across subcarriers, and indicate regions (values of f_k) where constructive or destructive interference occurs (frequency-selective fading).

- (d) If the number of subcarriers is halved to $N = 32$, recompute the subcarrier spacing. Discuss the resulting trade-off between higher subcarrier spacing (shorter symbol duration) and robustness to multipath delay spread.

Question 6: OFDM Design Tradeoffs

Consider an OFDM system with occupied bandwidth $B = 5$ MHz, number of subcarriers $N = 512$. Maximum channel delay spread $T_d = 4 \mu\text{s}$. Carrier frequency $f_c = 2.4$ GHz. Assume no guard bands or DC null; sampling rate $f_s = B$ and all N subcarriers carry data.

- For a *single-carrier* system occupying the same bandwidth B , what is the symbol time T_{SC} ?
- For the OFDM system, find the sample time T_s and the *raw* (useful) symbol time T_u (i.e., without CP).
- What is the *minimum* number of CP samples $N_{\text{CP,min}}$ required? Also report the corresponding minimum CP time $T_{\text{CP,min}}$.
- If the CP length is chosen as twice the minimum, i.e., $T_{\text{CP}} = 2T_{\text{CP,min}}$, what is the *total* OFDM symbol time $T_{\text{sym}} = T_u + T_{\text{CP}}$?
- What is the efficiency loss in percentage due to CP overhead?
- If 16-QAM is employed and all subcarriers carry data, what is the *effective* bit rate of this OFDM system (while accounting for CP overhead)?
- (ECSE 6560 only) Given $f_c = 2.4$ GHz, what is the *maximum mobile speed* v_{max} such that ICI remains negligible under the criterion $f_D \leq 0.1 \Delta f$?

Question 7: OFDM Bandwidth, Sampling Rate, and Aliasing

A real-valued OFDM signal occupies a total (two-sided) bandwidth of $B = 8$ MHz after digital upconversion to passband. The transmitter DAC operates at a sampling rate of $f_s = 20$ MHz, and the receiver ADC also samples at $f_s = 20$ MHz with ideal anti-alias filtering.

- State the Nyquist sampling condition to avoid aliasing. Does $f_s = 20$ MHz satisfy the condition for this signal? What is the highest frequency component (f_{max}) that can be perfectly reconstructed in this case?
- If the receiver mistakenly samples at $f'_s = 12$ MHz while keeping the same 8 MHz signal bandwidth, determine whether aliasing occurs. If it does, compute which frequency band of the transmitted signal (in MHz) would fold back into baseband.
- Draw a simple spectrum sketch showing: (i) the original 8 MHz-wide signal, (ii) the sampling replicas at $\pm f_s$, and (iii) what happens when the sampling rate drops to 12 MHz. Label the Nyquist limits and folded regions clearly.
- In practical OFDM systems, why is it preferable to leave a small "guard margin" between the occupied bandwidth and the Nyquist edge? Relate this to analog reconstruction filter imperfections and clock frequency offsets.

Question 8 (ECSE 6560 only): Residual CFO and SFO Estimation in OFDM

An OFDM system uses $N = 64$ subcarriers, sampling rate $f_s = 20$ MHz, and cyclic prefix length $N_{CP} = 16$. Two consecutive OFDM data symbols (after coarse synchronization and CFO correction) exhibit the following residual phase offset between their pilots:

$$\Delta\phi(f_k) = 2\pi f_k(N + N_{CP})\delta T_s - 2\pi\delta f_c(N + N_{CP})T_s,$$

where δf_c is the residual carrier frequency offset (CFO) and δT_s is the residual sampling period error (SFO).

- Explain how residual CFO and SFO manifest as a *constant* and a *linear* phase term across subcarriers. Sketch (or describe) the expected pilot phase vs. subcarrier index curve and identify which component corresponds to the CFO and which to the SFO.
- Given the following measured pilot phase differences between two consecutive OFDM symbols:

Subcarrier index (k)	$\Delta\phi_k$ [radians]
-26	+1.90
-13	+1.40
0	+0.90
+13	+0.40
+26	-0.10

Use linear regression $\Delta\phi_k = af_k + b$ to estimate the slope (a) and intercept (b). Take the subcarrier spacing $\Delta f = f_s/N = 312.5$ kHz.

- Compute the residual SFO δT_s and residual CFO δf_c in absolute Hz using:

$$a = 2\pi(N + N_{CP})\delta T_s, \quad b = -2\pi\delta f_c(N + N_{CP})T_s,$$

where $T_s = 1/f_s$. Express δT_s as a fractional error (ppm).

- Intuitively describe how residual CFO and SFO affect the constellation across OFDM symbols:
 - What visible distortion would residual CFO cause on a constellation diagram?
 - How would residual SFO appear over time if left uncorrected?